

A Review of the Generalized Black-Scholes Formula & It's Application to Different Underlying Assets

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Abstract

The Black-Scholes (1973) formula is well used for pricing vanilla European options. There are several different variations used by market practitioners dependent on the underlying asset being modelled. In this brief paper we present the generalized Black-Scholes representation and review how to configure the model appropriately for different asset classes. In particular varying the cost of carry term incorporated in the model allows us to express the generalized Black-Scholes as the classical Black-Scholes (1973) formula or the canonical Black (1976) representation.

Notation

The notation in table (1) will be used for pricing formulae.

C	Value of a European call option
b	The cost of carry, $b = r - q$
K	The strike of the European option
$N(z)$	The value of the Cumulative Standard Normal Distribution
ϕ	A call or put indicator function, 1 represents a call and -1 a put option
q	The continuous dividend yield or convenience yield
r	The risk-free interest rate (zero rate)
S	The underlying spot value
σ	The volatility of the underlying asset
T	The time to expiry of the option in years
P	Value of a European put option
V	Value of a European call or put option

Table 1: Notation

Introduction

The Black-Scholes (1973) model from [3] can be "generalized" by incorporating a cost of carry rate b . For an appropriate carry value the model can be used to price vanilla European options on stocks, stock indices paying a continuous dividend yield q , options on futures, commodity options incorporating the storage costs via a convenience yield q and currency options. The model's popularity amongst traders and market practitioners heavily relies on dynamic delta hedging, see [6] for details.

1 The Generalized Black-Scholes Formula

The generalized Black-Scholes (1973) formula is presented as

$$V = \phi e^{-rT} [S e^{bT} N(\phi d_1) - K N(\phi d_2)] \quad (1)$$

where

$$d_1 = \frac{\ln(S/K) + (b + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

for a call option setting $\phi = 1$ this becomes

$$C = e^{-rT} [S e^{bT} N(d_1) - K N(d_2)] \quad (2)$$

and likewise setting $\phi = -1$ for a put option we have

$$P = e^{-rT} [K N(-d_2) - S e^{bT} N(-d_1)] \quad (3)$$

Put-Call Super-Symmetry

The generalized formula (1) relies on Put-Call super-symmetry see [1] and [10], which also discuss negative volatility. Put-Call Super-Symmetry is a no-arbitrage condition whereby the price of a Call option C equals the negative price of an equivalent Put option P with negative volatility *ceteris paribus* namely

$$C(S, K, \sigma, T, r, b) = -P(S, K, -\sigma, T, r, b) \quad (4)$$

and equivalently

$$P(S, K, \sigma, T, r, b) = -C(S, K, -\sigma, T, r, b) \quad (5)$$

Cost of Carry

The cost of carry term b conveniently provides a mechanism to switch between the classical Black-Scholes (1973) and the Black (1976) formulas. Setting $b = r$ gives the Black-Scholes formula and likewise $b = 0$ the Black-76 formula as shown below.

2 The Black-Scholes (1973) Formula

The Black-Scholes (1973) formula from [3] is used for European stock options paying no dividends. Setting $b = r$ in the generalized Black-Scholes formula (1) transforms the pricing function into the classical Black-Scholes (1973) pricing kernel as shown below

$$V = \phi [SN(\phi d_1) - Ke^{-rT}N(\phi d_2)] \quad (6)$$

where

$$d_1 = \frac{\ln(S/K) + (r + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

3 The Black (1976) Formula

The Black (1976) formula from [4] is used for European options on forwards and futures. It is useful for interest rate cap and floor pricing for example. In a similar manner to the above, setting $b = 0$ in the generalized Black-Scholes formula (1) transforms the pricing function into the Black (1976) pricing formula as shown below

$$V = \phi [Se^{-rT}N(\phi d_1) - Ke^{-rT}N(\phi d_2)] \quad (7)$$

where

$$d_1 = \frac{\ln(S/K) + \frac{1}{2}\sigma^2T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

4 Put-Call Parity

Put-Call parity is well known no-arbitrage relationship between call and put options see [8] and [14]. A trading arbitrage opportunity exists if the parity relationship does not hold, subject to several dynamic hedging and trade replication assumptions namely

- No bid-ask trading spreads
- The market is frictionless i.e. no transaction costs
- The underlying asset can be traded long or short
- There are no liquidity restrictions on trade size

Mathematically we can express put-call parity as follows

$$C - P = Se^{(b-r)T} - Ke^{-rT} \quad (8)$$

or equivalently in terms of dividend yield q

$$C - P = Se^{-qT} - Ke^{-rT} \quad (9)$$

5 Applications of the Generalized Formula

The generalized Black-Scholes model is applied to different underlying assets by configuring the cost of carry term b as follows.

5.1 Black-Scholes Model (1973) - Stock Options

To price European stock options with no dividends we set $b = r$ whereby the generalized model represents the Black-Scholes (1973) model namely

$$V = \phi [SN(\phi d_1) - Ke^{-rT}N(\phi d_2)] \quad (10)$$

where

$$d_1 = \frac{\ln(S/K) + (r + \sigma^2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

5.2 Merton Model (1973) - Commodity Options, Stock Indices & Stocks with Dividends

To price European options with continuous dividends we set $b = r - q$ which leads to an expression for the Merton (1973) model see [12]. This is particularly useful for Commodity option pricing, whereby the q parameter incorporates storage and convenience yield costs. Likewise options on stock indices and stocks modelled having continuous dividends can be priced.

$$V = \phi [Se^{-qT}N(\phi d_1) - Ke^{-rT}N(\phi d_2)] \quad (11)$$

where

$$d_1 = \frac{\ln(S/K) + (r - q + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

5.3 Black Model (1976) - Options on Futures

To price European options on Future we set $b = 0$ and define the futures' price $F = e^{bT}$. The generalized Black-Sholes formula becomes

$$V = \phi [Fe^{-rT}N(\phi d_1) - Ke^{-rT}N(\phi d_2)] \quad (12)$$

where

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

5.4 Asay (1982) - Options on Margined Futures

To price European options on Futures, where the premium is paid into a margin account, we set both $b = 0$ and $r = 0$ and define the futures' price $F = e^{bT}$, see [2]. The generalized Black-Sholes formula becomes

$$V = \phi [FN(\phi d_1) - KN(\phi d_2)] \quad (13)$$

where

$$d_1 = \frac{\ln(F/K) + \frac{1}{2}\sigma^2T}{\sigma\sqrt{T}}$$

$$d_2 = d_1 - \sigma\sqrt{T}$$

5.5 Garman & Kohlhagen (1983) - Currency Options

To price Currency options see [7] we set $b = r_d - r_f$, $r = r_f$ and $q = r_d$ where r_d denotes the domestic (asset) interest rate and r_f the foreign (money) interest rate

$$V = \phi [S e^{-r_f T} N(\phi d_1) - K e^{-r_d T} N(\phi d_2)] \quad (14)$$

where

$$d_1 = \frac{\ln(S/K) + (r_d - r_f + \frac{1}{2}\sigma^2)T}{\sigma\sqrt{T}}$$
$$d_2 = d_1 - \sigma\sqrt{T}$$

6 Conclusion

In conclusion we have presented the generalized Black-Scholes formula and reviewed how by appropriately setting the cost of carry and interest terms we can apply the generalized result to price European options from different asset classes such as stock, commodity, futures and currency options.

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